QUALITY INFORMATION DOCUMENT

For OMI_CLIMATE_SI_ARCTIC_transport

Issue: 3.2

Contributors : Roshin P. Raj, Vidar S. Lien, Laurent Bertino

Approval Date : March 2026

MERCATOR OCEAN
INTERNATIONAL
2 avenue de l'Aérodrome de Montaudran, 31400 Toulouse, FRANCE
Tél : +33 5 61 39 38 02 - Fax : +33 5 61 39 38 99
Société civile de droit français au capital de 2 000 000 € - 522 911 577 RCS Toulouse - SIRET 522 911 577 00024

marine.copernicus.eu
mercator-ocean.eu

RECORD TABLE

When the quality of the products changes, the QuID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

Issue	Date	§	Description of Change	Author	Validated By
1.0	2021/01/10	All	Creation of the document	R. P. Raj, V. S. Lien	L. Bertino
2.0	2021/09/10	All	Document update	R. P. Raj, V. S. Lien	L. Bertino
3.0	2022/09/08	All	Document update	R. P. Raj, V. S. Lien	L. Bertino
3.2	2025/09/08	All	Document update	R. P. Raj, V. S. Lien	L. Bertino

TABLE OF CONTENTS

RECORD TABLE **2**

1) INTRODUCTION **4**

a) Product Name 4

b) Indicator Evaluation Method 4

c) Copernicus Marine products used as input for the computation 5

2) SCIENTIFIC VALIDATION **6**

a) Sea-ice transport 6

3) REFERENCES **8**

1) INTRODUCTION

a) Product Name

OMI_CLIMATE_SI_ARCTIC_transport

b) Indicator Evaluation Method

The time series provided here represents:

- Southward integrated sea-ice area and volume transport through the entire Fram Strait between Spitsbergen and Greenland
- Southward integrated sea-ice area and volume transport into the northern Barents Sea between Svalbard and Franz Josef Land
- Westward integrated sea-ice area and volume transport into the eastern Barents Sea between Novaya Zemlya and Franz Josef Land

The sea-ice area export (km^2 $month^{-1}$) is calculated as the sum of the width of each model grid cell times the sea-ice concentration within each grid cell times the sea-ice velocity perpendicular to the section. The sea-ice volume transport (km^3 $month^{-1}$) is calculated similarly, but also includes each grid cell's average sea-ice thickness. The time series of modelled sea-ice area and volume transport are evaluated against observation-based estimates reported in literature, assessing both the long-term averages and trends, and the seasonal variations.

c) Copernicus Marine products used as input for the computation

Product name (and Type)	Documentation
ARCTIC_MULTIYEAR_PHY_002_003	QUID: https://documentation.marine.copernicus.eu/QUID/CMEMS-ARC-QUID-002-003.pdf PUM: https://documentation.marine.copernicus.eu/PUM/CMEMS-ARC-PUM-002-003.pdf
ARCTIC_ANALYSISFORECAST_PHY_002_001	QUID: https://documentation.marine.copernicus.eu/QUID/CMEMS-ARC-QUID-002-001.pdf PUM: https://documentation.marine.copernicus.eu/PUM/CMEMS-ARC-PUM-002-001.pdf

2) SCIENTIFIC VALIDATION

a) Sea-ice transport

The sea-ice transports are calculated across three different sections, across the Fram Strait, and across two sections closing the Barents Sea to the north and east (see Fig. 1). The average monthly modelled sea ice export, both volume and area, follows the observations (e.g., Bi et al., 2016; Ricker et al., 2018) in terms of seasonal cycle, with a maximum in winter/spring (Nov-Apr) and a minimum in summer (Jul-Aug). The modelled annual average (1004×10^3 km^2) is around 50% higher than the estimate of 644×10^3 km^2 of Bi et al., 2016, which agrees with the overestimation of ice drift in the Fram Strait visible in Xie et al., 2017 (their Figure 14). When comparing with other models, the discrepancy is not unusual (Chevallier et al. 2017, Figure 10). Smedsrud et al. (2017) reported a tendency of a positive trend in sea-ice area transport during the most recent decades. However, the modelled area transport shows a negative trend of -0.4×10^3 km^2/year for the period 1993-2024 (Figure 2).
The long-term averaged annual modelled sea-ice volume transport of 1564 km^3 is higher than the observation-based estimate of 1132 km^3 reported by Zhang et al. (2017) for the period 1979-2012. The range of the modelled monthly maximum (415 km^3) and minimum (-17 km^3) sea-ice volume transport is comparable to the range of observation-based maximum (540 km^3) and minimum (21 km^3) sea-ice volume transport reported by Ricker et al. (2018).

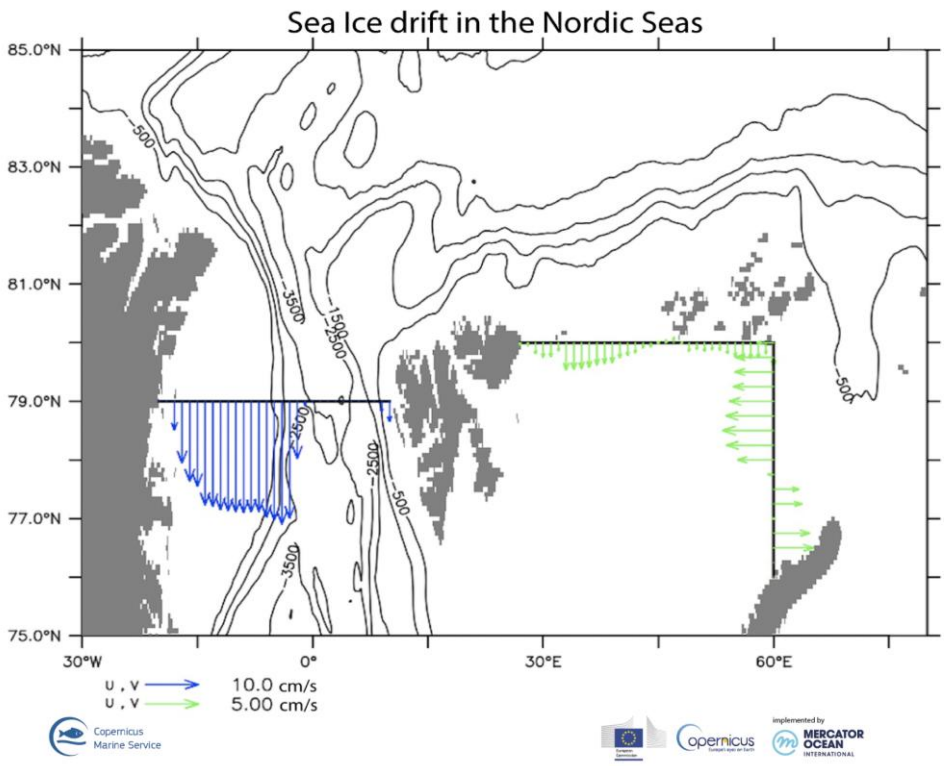

Figure 1: *Map showing the location of the sections across the Fram Strait between Greenland and Spitsbergen, the Barents Sea north between Svalbard and Franz Josef Land, and the Barents Sea east between Franz Josef Land and Novaya Zemlya, used to estimate the area and volume of sea ice transport. Mean sea ice drift velocity (perpendicular to the section)*

during the time period 1993-2021 is overlaid in arrows. Black isobaths are drawn at 500-meter interval.

Barents Sea

Less data and literature exist on the sea-ice transport from the Arctic Ocean to the Barents Sea. The modelled sea-ice area transport into the Barents Sea (both sections combined) show a small negative trend. Notably Lind et al. (2018), reported a strong decline in the sea-ice area import to the Barents Sea from 2000s until 2016. The reason for this discrepancy is not clear but points to deficiencies of sea ice drift. The modelled sea-ice volume transport to the Barents Sea, again both sections combined, shows linear trend of -0.3 km³/year for the period 1993-2024.

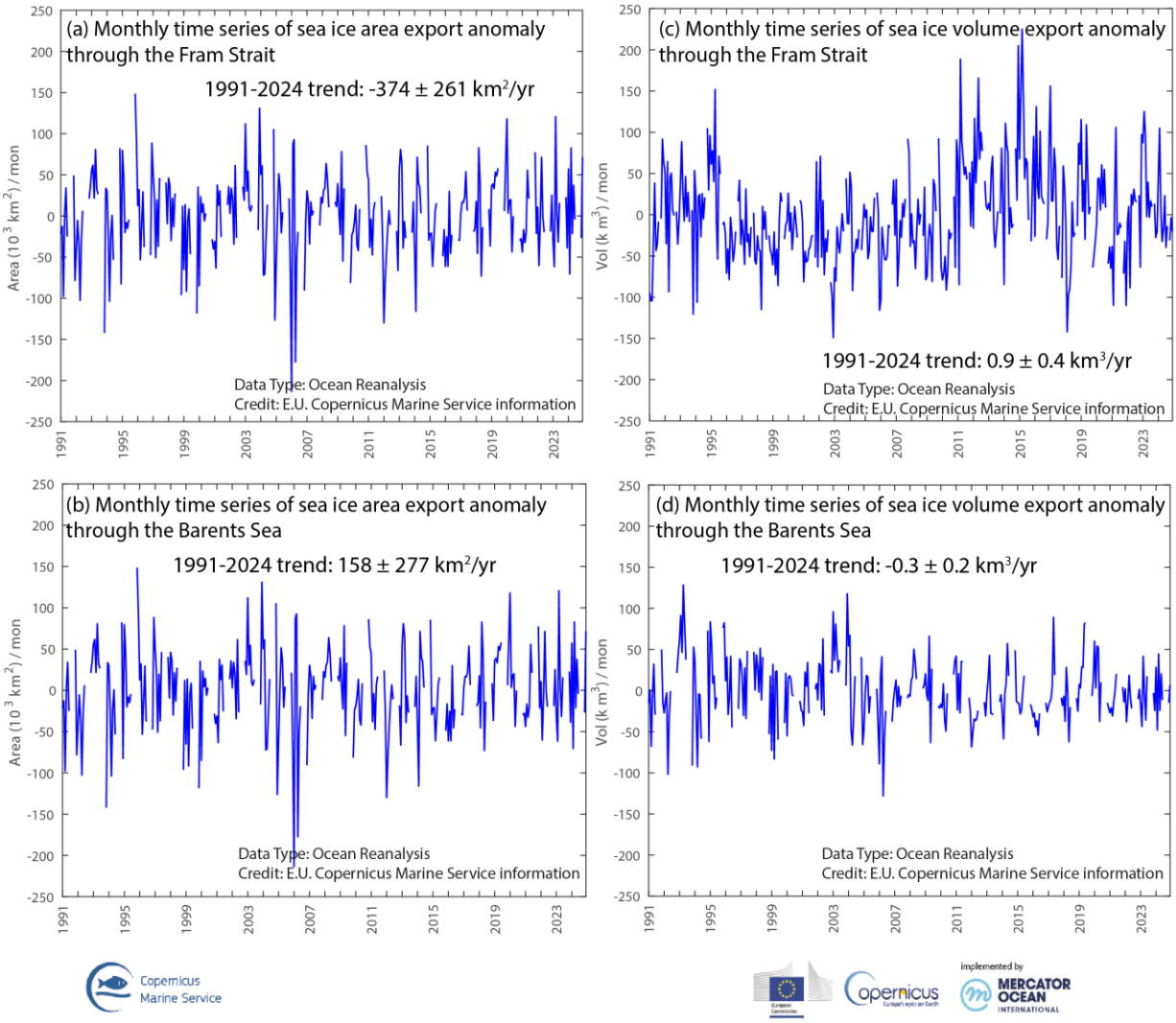

Figure 2: *Time series showing modelled monthly average sea-ice area (left) and volume (right) transport anomaly (relative to the mean seasonal cycle, 1991-2024) through the Fram Strait and Barents Sea.* Note the different units for area export in square kilometers (left) and volume export in cubic kilometers (right).

3) REFERENCES

Bi, H., Sun, K., Zhou, X., Huang, H., & Xu, X. (2016). Arctic Sea Ice Area Export Through the Fram Strait Estimated From Satellite-Based Data: 1988-2012. IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, 9(7), 3144-3157

Chevallier, M., Smith, G. C., Dupont, F., Lemieux, J. F., Forget, G., Fujii, Y., Hernandez, F., Msadek, R., Peterson, K. A., Storto, A., Toyoda, T., Valdivieso, M., Vernieres, G., Zuo, H., Balmaseda, M., Chang, Y. S., Ferry, N., Garric, G., Haines, K., Keeley, S., Kovach, R. M., Kuragano, T., Masina, S., Tang, Y., Tsujino, H. and Wang, X.: Erratum to: Intercomparison of the Arctic sea ice cover in global ocean–sea ice reanalyses from the ORA-IP project (Clim Dyn, 10.1007/s00382-016-2985-y), Clim. Dyn., 49(3), 1137–1138, doi:10.1007/s00382-016-3155-y, 2017.

Lind, S., Ingvaldsen, R.B., & Furevik, T. (2018). Arctic warming hotspot in the northern Barents Sea linked to declining sea ice import. Nature Climate Change, doi:10.1038/s41558-018-0205-y

Ricker, R., Girard-Ardhuin, F., Krumpen, T., & Lique, C. (2018). Satellite-derived sea ice export and its impact on Arctic ice mass balance. The Cryosphere, 12, 3017-3032

Smedsrud, L.H., Halvorsen, M.H., Stroeve, J.C., Zhang, R., & Kloster, K. (2017). Fram Strait sea ice export variability and September Arctic sea ice extent over the last 80 years. Cryosphere, 11(1), 65-79

Xie, J., Bertino, L., Counillon, F., Lisæter, K. A. and Sakov, P.: Quality assessment of the TOPAZ4 reanalysis in the Arctic over the period 1991–2013, Ocean Sci., 13(1), 123–144, doi:10.5194/os-13-123-2017, 2017.

Zhang, Z.H., Bi, H.B., Sun, K., Huang, H.J., Liu, Y.X., & Yan, L.W. (2017). Arctic sea ice volume export through the Fram Strait from combined satellite and model data: 1979-2012.